Probability and Statistics

Chapter 3. Random Variables and Probability Distributions

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Spring, 2026

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Chapter Overview

Concept of a Random Variable

Discrete Probability Distribution

Continuous Probability Distributions

Joint Probability Distributions

Concept of a Random Variable

Random Variables and Probability Distributions

A **random variable** translates outcomes in a sample space into real numbers.

This chapter studies:

- the concept of a random variable,
- discrete probability distributions,
- continuous probability distributions,
- joint probability distributions.

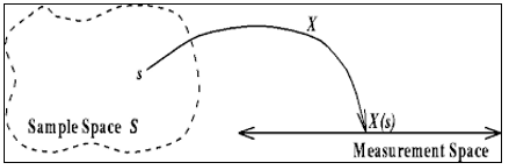
A random variable allows us to move from qualitative outcomes to numerical analysis.

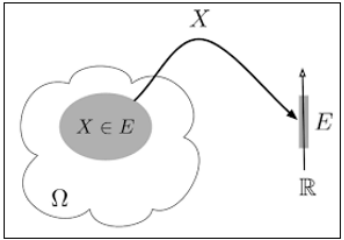
Definition of a Random Variable

Definition 3.1

A random variable is a function that associates a real number with each element in the sample space.

Illustration of a Random Variable





Discrete Sample Space

Definition 3.2

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**.

Continuous Sample Space

Definition 3.3

If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

Discrete Probability Distribution

Discrete Probability Distribution

A discrete random variable takes values in a finite or countably infinite set.

To describe such a variable completely, we specify

$$f(x) = P(X = x).$$

Probability Mass Function

Definition 3.4

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X, if for each possible outcome x,

$$f(x) \geq 0, \qquad \sum_x f(x) = 1, \qquad P(X = x) = f(x).$$

Cumulative Distribution Function for a Discrete RV

Definition 3.5

The cumulative distribution function $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \qquad -\infty < x < \infty.$$

Example 3.9

If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold.

Let X be the number of cars with side airbags among the next 4 cars sold.

Example 3.9 (Solution)

Each sale is independent and

$$P(\text{airbag}) = \frac{1}{2}.$$

For $x = 0, 1, 2, 3, 4$, the number of ways to obtain x cars with airbags is $\binom{4}{x}$. Hence,

$$f(x) = P(X = x) = \binom{4}{x}\left(\frac{1}{2}\right)^4, \qquad x = 0, 1, 2, 3, 4.$$

So $X \sim \text{Binomial}(4, \frac{1}{2})$.

Enumerating All Cases for Example 3.9

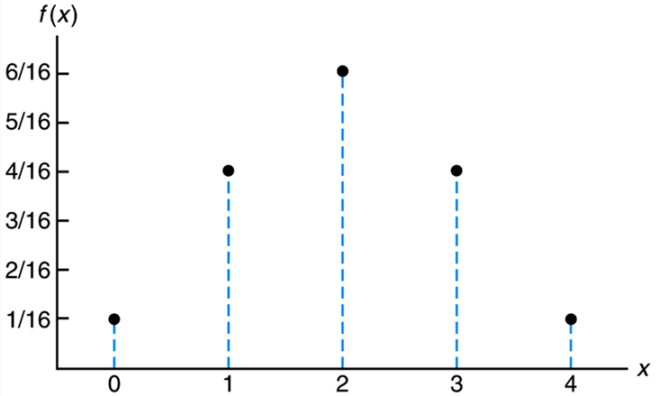
For intuition:

- $x = 0$: NNNN
- $x = 1$: ANNN, NANN, NNAN, NNNA
- $x = 2$: AANN, ANAN, ANNA, NAAN, NANA, NNAA
- $x = 3$: AAAN, AANA, ANAA, NAAA
- $x = 4$: AAAA

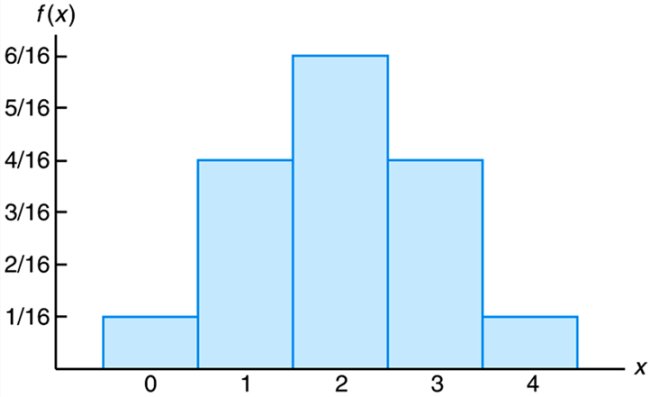
Thus, the number of outcomes is $\binom{4}{x}$, and

$$f(x) = \binom{4}{x} \left(\frac{1}{2}\right)^4.$$

Probability Mass Function Plot



Probability Histogram



Example 3.10

Find the cumulative distribution function of the random variable X in Example 3.9. Using $F(x)$, verify that $f(2) = 3/8$.

Sol)

From Example 3.9,

$$f(0) = \frac{1}{16}, \quad f(1) = \frac{4}{16}, \quad f(2) = \frac{6}{16}, \quad f(3) = \frac{4}{16}, \quad f(4) = \frac{1}{16}.$$

Example 3.10 (Solution)

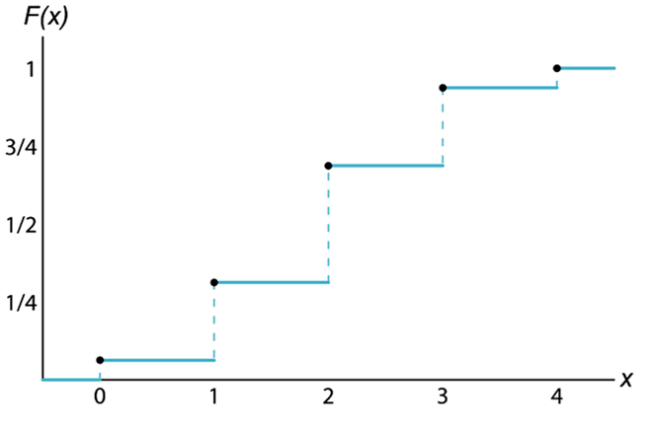
Therefore,

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{16}, & 0 \leq x < 1, \\ \frac{5}{16}, & 1 \leq x < 2, \\ \frac{11}{16}, & 2 \leq x < 3, \\ \frac{15}{16}, & 3 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

Then

$$f(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{6}{16} = \frac{3}{8}.$$

Discrete Cumulative Distribution Function



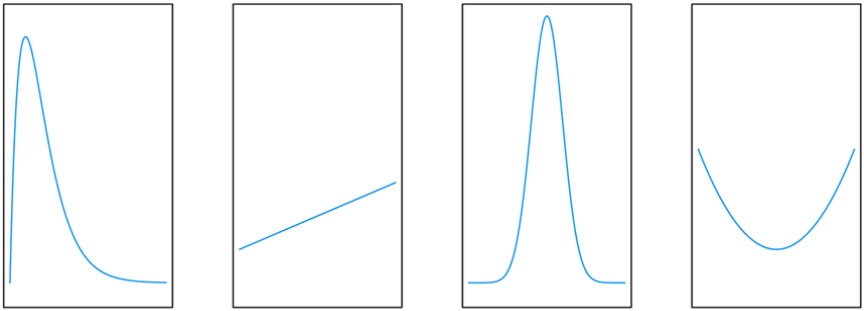
Continuous Probability Distributions

Continuous Probability Distributions

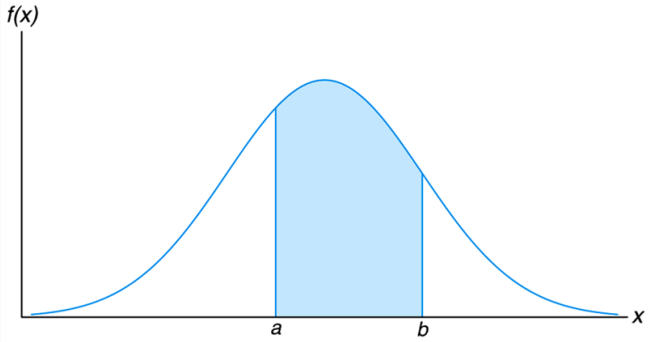
A continuous random variable takes values on an interval or collection of intervals.

For a continuous random variable, probabilities are described by a **density function**, and probabilities correspond to **areas under a curve**.

Typical Density Functions



Interpreting $P(a < X < b)$



Probability Density Function

Definition 3.6

The function $f(x)$ is a probability density function (pdf) for the continuous random variable X, defined over the set of real numbers, if

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x)\,dx = 1,$$

and

$$P(a < X < b) = \int_a^b f(x)\,dx.$$

Example 3.11

Suppose that the error in the reaction temperature, in ℃, for a controlled laboratory experiment is a continuous random variable X having the density

$$f(x) = \begin{cases} \dfrac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- Verify that $f(x)$ is a density function.
- Find $P(0 < X \leq 1)$.

Example 3.11 (Solution)

Since $f(x) = x^2/3 \geq 0$ on $(-1, 2)$, we only need to verify the total area:

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-1}^{2} \frac{x^2}{3}\,dx = \left[\frac{x^3}{9}\right]_{-1}^{2} = \frac{8}{9} + \frac{1}{9} = 1.$$

Hence $f(x)$ is a pdf.

Also,

$$P(0 < X \leq 1) = \int_{0}^{1} \frac{x^2}{3}\,dx = \left[\frac{x^3}{9}\right]_{0}^{1} = \frac{1}{9}.$$

Cumulative Distribution Function for a Continuous RV

Definition 3.7

The cumulative distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt, \qquad -\infty < x < \infty.$$

Example 3.12

Find the cumulative distribution function $F(x)$ for the density in Example 3.11.

Sol) For the pdf

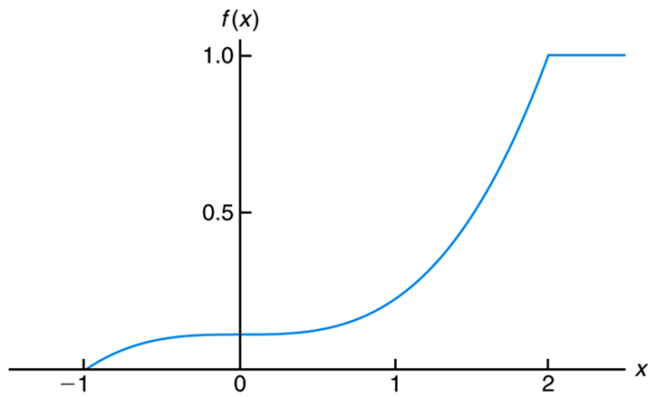
$$f(x) = \begin{cases} \dfrac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere}, \end{cases}$$

Example 3.12 (Solution)

the cdf is

$$F(x) = \begin{cases} 0, & x \leq -1, \\ \displaystyle\int_{-1}^{x} \frac{t^2}{3}\, dt = \frac{x^3+1}{9}, & -1 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

Continuous Cumulative Distribution Function



Joint Probability Distributions

Joint Probability Distributions

So far we have considered one random variable.

In many applications, two or more random variables are observed together. This leads to **joint probability distributions**.

Joint Probability Distribution of Discrete Variables

Definition 3.8

Let X and Y be discrete random variables. The function

$$f(x, y) = P(X = x,\ Y = y)$$

is called the **joint probability distribution** of X and Y, provided

$$f(x, y) \geq 0, \qquad \sum_x \sum_y f(x, y) = 1.$$

Example 3.14

Consider two discrete random variables X and Y with joint probability distribution given in Table 3.1.

This table will be used to illustrate:

- marginal distributions,
- conditional distributions,
- independence.

Table 3.1: Joint Probability Distribution

	$f(x,y)$	x = 0	x = 1	x = 2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Marginal Distribution of X

Definition 3.9

The marginal distribution of X is obtained by summing the joint distribution over all values of Y:

$$f_X(x) = \sum_y f(x, y).$$

Marginal Distribution of Y

Definition 3.10

The marginal distribution of Y is obtained by summing the joint distribution over all values of X:

$$f_Y(y) = \sum_x f(x, y).$$

Joint Distribution of Continuous Variables

Definition 3.11

If X and Y are continuous random variables, their joint density function $f(x, y)$ satisfies

$$f(x, y) \geq 0, \qquad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1.$$

Marginal Densities

Definition 3.12

The marginal density functions are

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx.$$

Independence via Joint Distributions

Definition 3.13

The random variables X and Y are independent if

$$f(x, y) = f_X(x) f_Y(y)$$

for all x, y.

For continuous random variables, the same factorization criterion applies to the densities.